

Practical No.7

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30-08-2025

Objective:

Let the lifetime of rabbits of a particular species follows exponential with rate parameter 0.3 . Find the probability that the randomly chosen rabbit of such species will:

- i) Survive at least for 10 days
- ii) Die before 20 Days
- iii) Will die with 15 to 25 Days

Python Code

```
In [6]: import numpy as np
def cdf_exp(l,t):
    prob=1-np.exp(-l*t)
    return prob
```

```
In [9]: 1-cdf_exp(0.3,10)
```

```
Out[9]: np.float64(0.04978706836786395)
```

```
In [10]: cdf_exp(0.3,20)
```

```
Out[10]: np.float64(0.9975212478233336)
```

```
In [12]: cdf_exp(0.3,25)-cdf_exp(0.3,15)
```

```
Out[12]: np.float64(0.010555912168094483)
```

Interpretation:

i) As we can see the the defined function 'cdf_exp' is giving values for surviving of rabbit at least for 10 days is 0.04978706836786395.

ii) As we can see the the defined function 'cdf_exp' is giving values for rabbit Die before 20 Days is 0.9975212478233336.

iii) As we can see the the defined function 'cdf_exp' is giving values for rabbit Die in between 15 to 20 Days is 0.010555912168094483